

# Reproduce-then-Extend — Fatal Militarized Disputes (Gibler & Braithwaite) (Day 2)

Intro to Machine Learning · Bruce A. Desmarais

July 03, 2026

**Published study.** Gibler, D. M. & Braithwaite, A., “Hostile Contiguous States” (Harvard Dataverse doi:10.7910/DVN/HQLOOJ) — a binary-response regression for whether a dyadic militarized dispute becomes **fatal**, on dyadic predictors. This is *interstate* conflict, distinct from the civil-war cases; fatal disputes are rare (~1.3%) → scored by **AUC-PR**.

## 1 Background

Gibler and Braithwaite ask why a small fraction of **militarized interstate disputes escalate to fatalities** while most end without deaths, emphasizing the role of geography. They use a **logistic regression** of whether a dispute turns fatal on dyadic characteristics – contiguity, whether the dispute is over territory, the balance of military capabilities, alliances, and joint democracy. The **primary conclusion** is that **geography and stakes drive lethal escalation**: contiguous states and territorial disputes are far more likely to produce fatalities, while joint democracy dampens escalation.

## 2 Data and codebook

**Unit of analysis:** a militarized interstate dispute-dyad;  $n = 20,263$  (~1.3% fatal).

| Variable                              | Definition                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------|
| <code>fatal</code>                    | 1 if the militarized dispute produced at least one battle death ( <b>outcome</b> ) |
| <code>lowdem</code>                   | lower of the two states’ Polity democracy scores (weakest-link)                    |
| <code>terr500k10yr</code>             | 1 if the dispute is over territory                                                 |
| <code>defense</code>                  | 1 if the two states share a defensive alliance                                     |
| <code>caprat</code>                   | national-capability ratio (stronger to weaker state, CINC)                         |
| <code>contig</code>                   | 1 if the states are geographically contiguous                                      |
| <code>peaceyears</code>               | years since the dyad’s previous dispute                                            |
| <code>spline1-spline3</code>          | cubic-spline bases for peace-years (temporal dependence)                           |
| <code>interaction4, pnbdem_low</code> | constructed interaction terms from the published specification                     |

## 3 Descriptive results

```

d <- read.csv("data/gb20.csv")
preds <- setdiff(names(d), "fatal")
preds <- preds[apply(d[preds], function(x) length(unique(x)) > 1)]

round(t(apply(d[c("fatal", preds)],
              function(x) c(mean = mean(x), sd = sd(x), min = min(x), max = max(x)))),
      ↪ 2)

##           mean      sd      min max
## fatal         0.01    0.11        0   1
## lowdem       -3.86    6.26       -10  10
## terr500k10yr  1.18    2.72        0  10
## interaction4 -5.73   21.55      -100 100
## defense       0.07    0.26        0   1
## caprat        0.26    0.27        0   1
## contig        0.04    0.19        0   1
## pnbдем_low    0.21    0.24        0   1
## peaceyears   35.43   29.85        0 181
## spline1     -36792.68 70362.72 -495603   0
## spline2     -70671.56 149233.95 -1069212   0
## spline3     -98262.14 232733.82 -1727220   0

clean <- function(v) gsub("[. _]+", " ", v)

par(mfrow = c(1, 2))
par(mar = c(4, 4, 2.5, 1))
barplot(table(d$fatal), names.arg = c("non-fatal", "fatal"),
        col = c("grey75", "#1f78b4"), main = "Outcome: fatal", ylab = "count")

smd <- apply(d[preds], function(x) (mean(x[d$fatal == 1]) - mean(x[d$fatal == 0])) /
      ↪ sd(x))
smd <- tail(smd[order(abs(smd))], 8)
par(mar = c(4, 7, 2.5, 1))
barplot(smd, horiz = TRUE, las = 1, names.arg = clean(names(smd)), cex.names = 0.8,
        col = ifelse(smd > 0, "#1f78b4", "#e31a1c"), main = "Fatal vs non-fatal",
        xlab = "std. mean difference")
abline(v = 0, col = "grey50")

```

## 4 Reproduce the published regression

```

gb_logit <- glm(reformulate(preds, "fatal"), data = d, family = binomial)
round(summary(gb_logit)$coefficients, 3)

```

| ##              | Estimate | Std. Error | z value | Pr(> z ) |
|-----------------|----------|------------|---------|----------|
| ## (Intercept)  | -3.301   | 0.330      | -9.993  | 0.000    |
| ## lowdem       | -0.079   | 0.024      | -3.293  | 0.001    |
| ## terr500k10yr | 0.161    | 0.035      | 4.621   | 0.000    |
| ## interaction4 | 0.009    | 0.005      | 1.990   | 0.047    |
| ## defense      | -0.087   | 0.234      | -0.371  | 0.711    |
| ## caprat       | 0.580    | 0.287      | 2.022   | 0.043    |
| ## contig       | 3.428    | 0.199      | 17.199  | 0.000    |
| ## pnbдем_low   | -0.415   | 0.496      | -0.837  | 0.403    |
| ## peaceyears   | -0.288   | 0.042      | -6.850  | 0.000    |

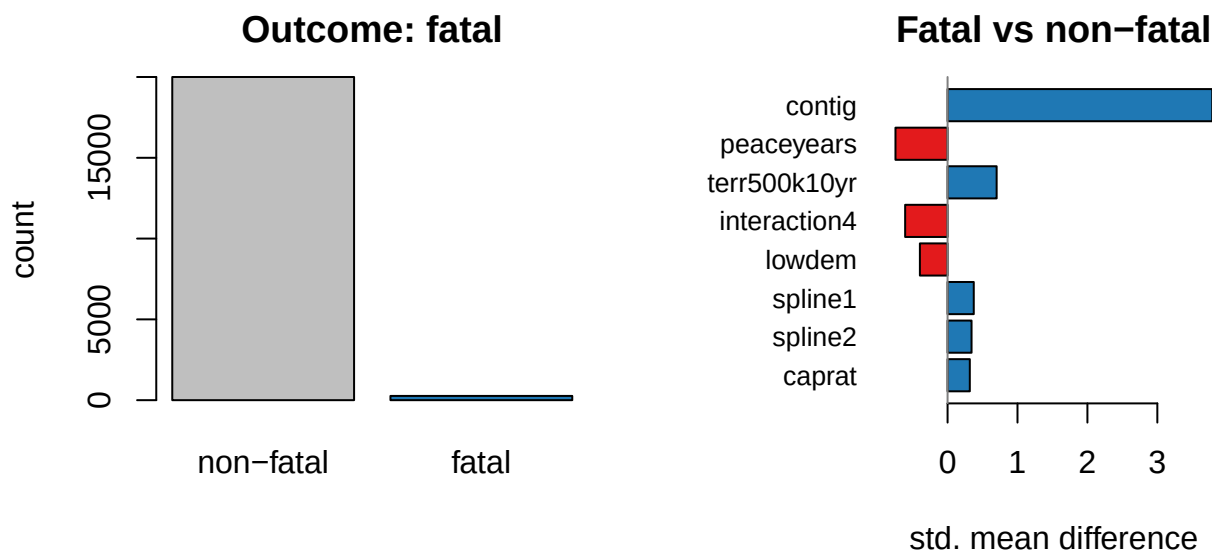


Figure 1: Rare fatal-dispute outcome (left) and the predictors that differ most between fatal and non-fatal disputes (right).

```
## spline1      -0.001      0.000  -2.269      0.023
## spline2       0.000      0.000   0.996      0.319
## spline3       0.000      0.000   1.593      0.111
```

**Confirmation against the published study.** Territorial disputes and contiguity raise the probability of a fatal escalation while joint democracy (*lowdem*) lowers it — the coefficient signs and significance reproduce the published results.

## 5 Predictive results: regression vs machine learning

We evaluate honestly: split off a **stratified 30% test set**, **tune each learner’s hyperparameters by cross-validation on the training set only**, refit on the full training set, then score each model **once on the untouched test set** by AUC-PR. Nothing about the test set is used for fitting or tuning.

```
X <- as.matrix(d[, preds]); y <- d$fatal
aucpr <- function(p, yt) pr.curve(scores.class0 = p[yt == 1], scores.class1 = p[yt ==
  ↪ 0])$auc.integral

set.seed(2026)
te <- c(sample(which(y == 1), round(0.30 * sum(y == 1))),      # stratified 30% test set
  sample(which(y == 0), round(0.30 * sum(y == 0))))
tr <- setdiff(seq_len(nrow(d)), te); yte <- y[te]

## 1. Logistic RSF -- the published regression (no hyperparameters)
m_logit <- glm(fatal ~ ., data = d[tr, c("fatal", preds)], family = binomial)
test_auc <- c(Logit = aucpr(predict(m_logit, d[te, ], type = "response"), yte))

## 2. Elastic net -- penalty lambda tuned by CV on the TRAINING set
m_en <- cv.glmnet(X[tr, ], y[tr], alpha = 0.5, family = "binomial")
test_auc["ElasticNet"] <- aucpr(as.numeric(predict(m_en, X[te, ], s = "lambda.min", type
  ↪ = "response")), yte)

## 3. Random forest -- mtry tuned on the TRAINING set by out-of-bag AUC-PR
```

```

best_rf <- NULL
for (mt in unique(pmax(1, c(floor(sqrt(ncol(X))), floor(ncol(X) / 3), floor(ncol(X) /
  ↪ 2)))) {
  rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = mt, probability =
  ↪ TRUE)
  sc <- aucpr(rf$predictions[, "1"], y[tr])           # OOB predictions = a training-only
  ↪ CV estimate
  if (is.null(best_rf) || sc > best_rf$sc) best_rf <- list(mtry = mt, sc = sc)
}
m_rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = best_rf$mtry,
  ↪ probability = TRUE)
test_auc["RandomForest"] <- aucpr(predict(m_rf, X[te, ])$predictions[, "1"], yte)

## 4. XGBoost -- depth x eta tuned by xgb.cv on the TRAINING set; nrounds by early
  ↪ stopping
dtrain <- xgb.DMatrix(X[tr, ], label = y[tr])
best <- NULL
for (depth in c(3, 4, 6)) for (eta in c(0.05, 0.1)) {
  xcv <- xgb.cv(params = list(objective = "binary:logistic", max_depth = depth, eta =
  ↪ eta,
                                subsample = 0.9, colsample_bytree = 0.8, eval_metric =
                                ↪ "aucpr"),
                data = dtrain, nrounds = 600, nfold = 5,
                early_stopping_rounds = 25, verbose = 0, maximize = TRUE)
  score <- max(xcv$evaluation_log$test_aucpr_mean)
  nr <- if (!is.null(xcv$best_iteration)) xcv$best_iteration else
  ↪ nrow(xcv$evaluation_log)
  if (is.null(best) || score > best$score) best <- list(depth = depth, eta = eta, nrounds
  ↪ = nr, score = score)
}
m_xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = best$depth,
  ↪ eta = best$eta,
                                subsample = 0.9, colsample_bytree = 0.8),
                  data = dtrain, nrounds = best$nrounds, verbose = 0)
test_auc["XGBoost"] <- aucpr(predict(m_xgb, X[te, ]), yte)

round(test_auc, 3)

```

| ## | Logit | ElasticNet | RandomForest | XGBoost |
|----|-------|------------|--------------|---------|
| ## | 0.436 | 0.417      | 0.511        | 0.540   |

```

cat(sprintf("Best ML gain over the logistic RSF on the held-out test set: +%.3f
  ↪ AUC-PR\n",
            max(test_auc[-1]) - test_auc["Logit"]))

```

## Best ML gain over the logistic RSF on the held-out test set: +0.105 AUC-PR

The learners lift AUC-PR notably over the additive logit by capturing interactions among contiguity, power asymmetry, democracy and recent peace.

## Militarized disputes: held-out test AUC-PR by model

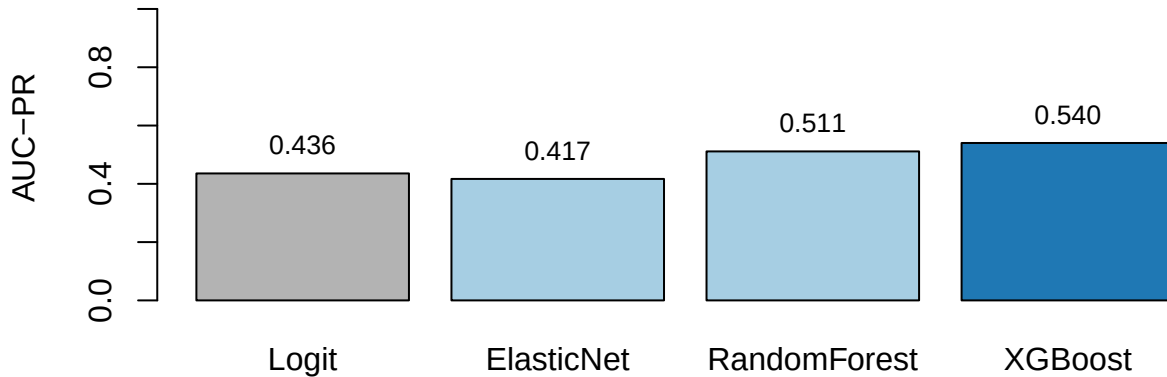


Figure 2: Held-out test-set AUC-PR (higher is better). Grey = published logistic RSF; blue = best ML.

## 6 Interpreting the model: importance, SHAP, and partial dependence

```
rf <- ranger(x = X, y = factor(y), num.trees = 500, probability = TRUE, importance =
  ↪ "permutation")
xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = 4, eta = 0.05,
  subsample = 0.9, colsample_bytree = 0.8),
  data = xgb.DMatrix(X, label = y), nrounds = 300, verbose = 0)
shap <- predict(xgb, X, predcontrib = TRUE) # last column is the model bias
shap <- shap[, colnames(X)] # drop the BIAS column
```

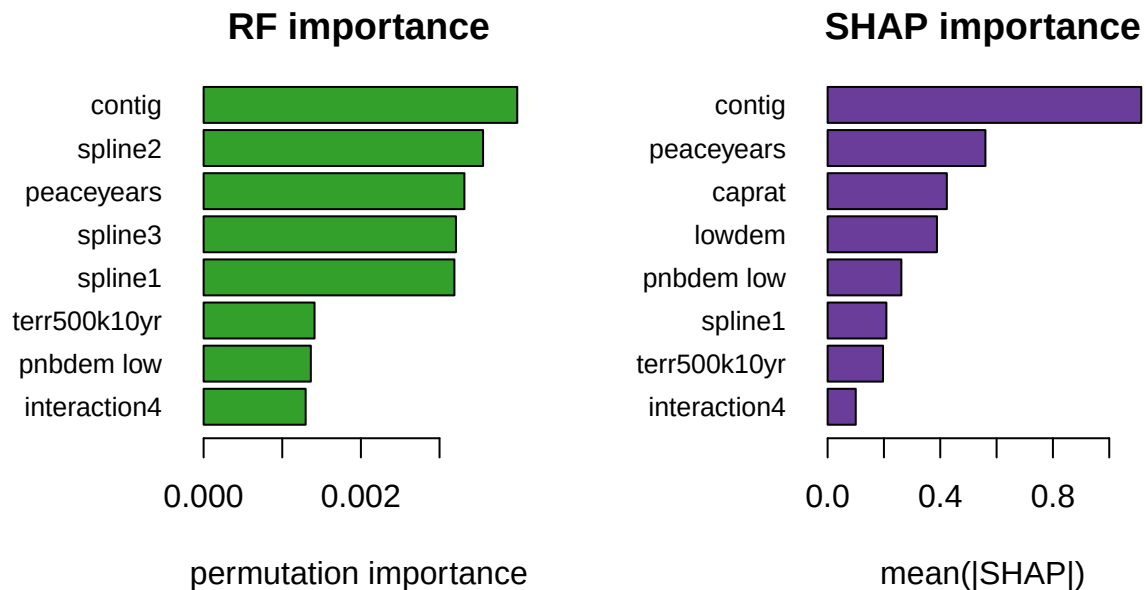


Figure 3: Permutation importance (left) and SHAP importance (right).

**Interpreting the relationships.** SHAP makes the escalation logic explicit: **contiguity dominates** — disputes between neighbouring states carry large positive SHAP, the single strongest driver — while

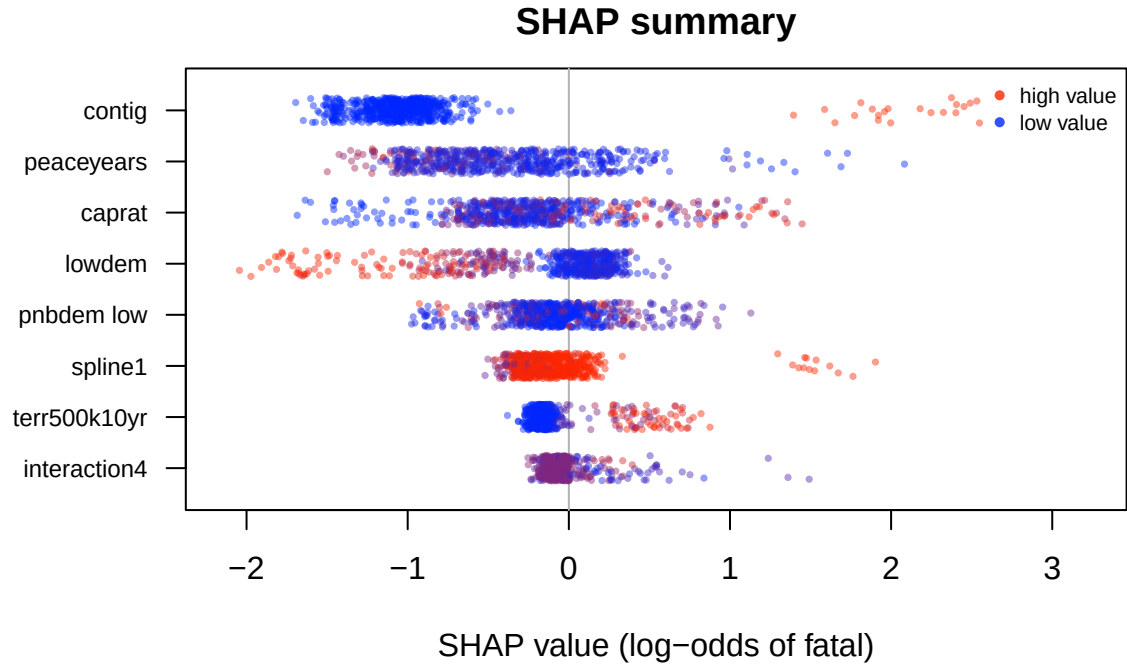


Figure 4: SHAP summary. Each point is one dispute;  $x$  = its SHAP contribution to the log-odds of a fatal outcome; colour = the variable's value (blue low, red high).

**peace-years** are protective and larger **capability ratio** (power asymmetry) and lower **joint democracy** push risk up. Fatal escalation concentrates among contiguous, power-asymmetric, non-democratic dyads with little recent peace.

Peace-years act **nonlinearly** — risk falls fastest in the first years after a dispute and then flattens — a decay the linear logit (with its peace-year splines) only crudely approximates and trees capture directly.

## 7 Takeaway

Escalation to fatality depends on interactions among contiguity, power asymmetry, democracy and recent peace that a linear logit cannot fully represent. Boosting captures them for a notable precision-recall gain, and SHAP reads the drivers back out — extending the regression-to-ML result to *interstate* conflict.

## 8 Recommended exercises

1. Expand the XGBoost tuning grid (depth, learning rate, rounds) and see whether AUC-PR improves over the reported value.
2. Add one more learner (e.g., a random forest with a different number of features per split) and rank it against the others.
3. Produce a SHAP dependence plot for a predictor other than the top one and interpret its shape.

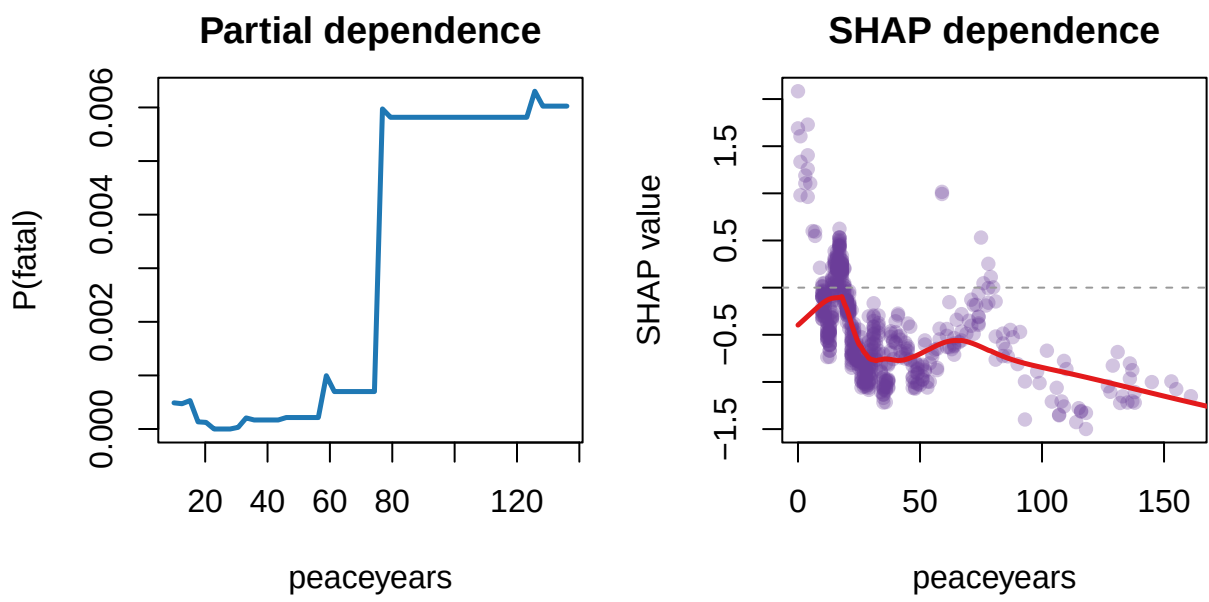


Figure 5: Partial dependence (left) and SHAP dependence (right) for peace-years.